



Toward a Systematic Methodology for Developing Business Performance Management Systems

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Abstract: Business performance management (BPM) systems involve many features different from traditional Enterprise resource planning (ERP) applications. Based on academic research and industrial experiences on developing BPM systems, this paper describes a framework and methodology for understanding the process of BPM design and implementation projects. The methodology was applied and proved in developing BPM systems for large companies in China. This paper reviews the researches of BPM system development and discusses the concept and the methodology for developing BPM systems. Based on the real case study, the research analyzes the characteristics and critical issues of the design and the implementing of BPM systems. It discusses the concept and the basic working principles of BPM systems. A conceptual framework of the methodology for developing BPM systems is built upon the study.

Keywords: business performance management, information systems, design process, system development

1 Introduction

The concepts and applications of Business performance management (BPM) have raised significant attention among researchers and practitioners in the recent years. According to a survey on the top-500 enterprises in the US by Business Weekly, corporations around the world are competitively adopting performance management strategies. With annual growth of 40% in the US BPM market, the expenses on BPM software development will increase in great scale [1]. In addition, three major BPM manufacturers: Hyperion, Business Object and Cognos were respectively purchased by Oracle, SAP and IBM, with the total amount in this acquisition deal up to \$13 billion. This means that a new kind of competition in BPM is on the way.

Chinese enterprises have two reasons to implement BPM system. One is that as their direct cooperation and competition with foreign enterprises, they have gradually realized the inherent problems in their operational modes. The other is that enterprises are more and more capable

of obtaining accurately their inner operational data with the development of information technology, and hence are considering how to improve their own ability in performance management. As a result, the introduction of BPM systems into the organizations has become one of the critical management problems. However, many companies lack methodologies for developing their BPM systems. BPM is still a new specific information system involved with knowledge and practice from multiple disciplines. To develop the unique BPM features cannot be resolved by simply using the methods in traditional ERP application. Based on our academic research and industrial experience on BPM systems, we developed a framework and methodology for understanding the process of BPM design and implementation projects. The methodology was applied and proved in developing BPM systems for large companies in China.

This paper has the following structure: the second part discusses briefly the concept and the basic working principles of BPM systems; the third part focuses on a case study by introducing the developmental process of BPM in a real project; a conceptual framework of the methodology for developing BPM systems is built upon the case in the fourth part; the final part of this paper draws a conclusion and discusses the future studies in this area.

2 Previous researches

There are various definitions of business performance management systems. Some enterprises, such as Hyperion name it BPM (Business Performance Management) [2], some eEnterprise, such as Oracle and BO, call it EPM (Enterprise Performance Management), and still some scholars put it PMS (Performance Management System) or CPM (Corporation Performance Management) [3,4]. Despite of the differences in naming, the objections processed by the system and the goals of the system are more or less the same when mentioned. The definition given by International Data Center (IDC) is that: BPM system is a cross-industry and cross-function interactive applied software, and it supports the management of business performance by measuring and analyzing the business performance

mainly according to a series consistent and identifiable Key Performance Index (KPI) [5]. The global Data Warehouse Research Institution (TDWI) has carried out an investigation on 675 persons among whom there are high-level managers whose enterprises have already implemented BPM system, system designers and users. It developed the following definition on the basis of their survey: BPM is a series of process and application system under serious consideration with the aim to optimize strategic execution. BPM process includes four steps: strategizing, business planning, monitoring and adjustment/acting. Each step is fulfilled by various tools of which the data warehouse and business intelligence are the most important [3]. The Business Performance Management and Implementation Association (BPMIA) defines BPM as follows:

(1) BPM is an integrated, internal-circulated, analytical and managerial system based on technology support.

(2) BPM is the enabler to define strategic targets, and to supervise the business performance according to the targets.

(3) The main processes in BPM include: business planning, preparation and reporting of the performance index data, business model, analysis and supervision on KPI and improvement in process.

From the perspective of information systems, unlike Balance Scorecard and employee performance evaluation mechanism, BPM is an information system to plan, evaluate, monitor and adjust the performance on the organizational level. In essence, the main function of BPM system focuses on specific management decisions based on the data and information accumulated among the different levels of organization. By improving business processes, BPM system provides the users with a set of feasible schemes to direct their decision process and ensure effective execution of the decision. As the main users of BPM systems are the high- to media-level managers in the enterprises, the main task of BPM systems is to help managers get control of the execution of the strategic target, and make timely adjustment according to the change of environment. Because of this, financial software becomes an important part of BPM. Take Oracle as an example: its BPM system has a transaction processing system whose function are related with sales, CRM, supply chain, financing, human resources, purchasing and so on; a business intelligent system including data warehouse, OLAP, inquiring analysis, data integration, decision control cabin and so on; and a financial system with functions in standards and evaluation, budget and planning, financial tables and financial modeling [6].

As discussed before, the nature of BPM information system are demonstrated in Table 1.

The studies on BPM system are relatively few in the academic field of information system, and most frequently quoted papers in the past three years are from some authorized academic journals, such as MISQ, JMIS, CAIS, and so on, with only a few papers on BPM. In

addition, there is no consensus on the methodology of building BPM systems to guide practices in system development and implementation. This phenomenon demonstrates a distinct contrast between the presently inadequate theoretical research and the growing demand in practice in the information systems area.

Tab.1 The nature of BPM information system

nature	explanation
users	managers of all levels in the enterprises especially top managers
function of the system	monitor and adjust the development and execution of enterprise strategies by put organizational performance into digital form
working object	organization, department and process
output of the system	performance information and KPI of concerned organization, department and working process
input of the system	data acquired through transaction processing system, usually with independent performance management database
interface	visual graphic demonstration, and control cabin-type users' interface
main tools	business intelligence, data warehouse, financial planning, balanced score cards (BSC), developing tools, etc.

One of the most important and practical problem in theoretical study of BPM lies in the fact that decision makers are faced with perplexities when their enterprises are adopting BPM. System architects and managers cannot see the differences between BPM systems and ERP systems. They doubt that the traditional methodology of ERP system development can be easily used to develop BPM system. This paper uses case based empirical study to form a referential framework for system developing methodology according to the experience gained through real BPM projects.

3 A case study on designing a BPM system

3.1 The background of BPM system development

CTV is a big state-owned media group of over ten thousand staffs, with a dozen of transaction department, companies and business units as its subordinates. It has been developed into an organization with various systems of ownership. In the recent few years, as the fast pace in economic growth in China, the media business has experienced extremely high-speed development. This group broadcast more and more programs and had a great social influence, which draws significant attention

from the TV-watchers, government and the society. In order to maintain its constant growth, the top managers urgently needed to get the accurate working performance information of each department, and meanwhile a right evaluation on the working state and performance of the subordinate institutions, to realize the potential problems in the group and make adjustment and changes in policies and resources correspondingly. Therefore, the top managers have long been paying great attention to BPM system implementation.

CTV had developed many legacy information systems and realized the net-based informational management of intra-organizational daily business, but the systems and databases are limited to the use of daily transaction processing. The subordinate departments of the group can produce a large amount of data everyday and give a huge burden to the performance management of top managers. In order to solve this problem, a specific investigation and study group was established to arrange the information received from the departments by direct communication and paper survey. However, such fixed hand-acquiring information obtaining process was affected by the time when the departments provide their information, which can hardly ensure that files can be timely presented to the top managers. The craft of the personnel in sorting out the information became essential in this process. A report usually required around a month to be prepared and could not be submitted to top managers with accurate decision data, and thus they cannot have precise information of the intra-organizational performance decision.

In order to solve this problem, CTV started a BPM project by cooperation with a university and a software company [7]. The object of this project was to develop a new BPM system to facilitate the managers to get timely information on the finance, market, program making and human resources of the group, and generate understandable knowledge of the overall operational situations. At the same time, a reasonably designed analytical model will assist the managers in getting the information of their operational performance. The developing team consists of researchers from the management school of Peking University (BPM researchers), technicians from a software company (IT technician) and personnel in a special field from relative departments in CTV (users). After analyzing the objectives and functions of the system, the team reached an agreement of the scope of work: BPM researchers were in charge of system analysis and design; IT technicians were in charge of data arrangement and system implementation; and users were responsible to the managerial tasks such as connection among the three parties, control the work progress and so on.

3.2 Systematic investigation and study, and business analysis

The team first had a thorough investigation of the organizational structure, business type and factors related to performance management in CTV, and after

interviews and business process analysis, they understood that the organizational structure of CTV was a pyramid with station general managers on top, and then the various centers (including news, abroad, economy, art & literature, sports, etc.), and direct offices (such as chief editor's office, HR office, financial office, etc.). At their subordinate level, there were three branch departments and divisions. The centers were the main institutions which took in charge of business operation such as TV program making and broadcasting. There were total 15 second-level institutions and 90 third-level institutions in this TV station. The main product of the station was TV programs and broadcasting services. The program making was monitored by the centers. There was also a three-level relationship in the broadcast programs from channel, to column, to show. Each show was broadcast to the audience through channels, and each channel focused on one topic, such as art & literature, news, etc. The station owned 15 channels in total. Each channel has over ten columns with each column was broadcasted in a fixed name and time, and the shows on the columns were the products of various program making centers.

3.3 Performance factors collection and KPI extraction

A key problem of BPM system designing is how to extract the needed performance factors. According to the requirements of clients, the system's users are identified as the leaders of the TV station and the assistant agent of decision-making (the investigation department of the direct subordinate agents), and the functions of senior executive supporting were defined as the following three areas:

(1) Strategic goal breakdown. The strategic goals of the TV station needed to be decomposed from several different perspectives. For example, the goals such as improving programs' ratings were distributed to the centers, and further assigned to the columns.

(2) The performance evaluation of subordinate organizations. By using this system, leaders could know the operation statuses of the subordinate organizations in time, so that the problems within the organizations could be detected quickly.

(3) Decision-supporting. By using a series of models provided in the system, users could operate on the data to investigate the organization's situation and conduct development trend analysis and forecasting.

We divided the performance indicators concerned by the top leaders into strategic indicators and process indicators. The strategic indicators were the indicators which were quantified by synthesizing indicators from several centers (including the three directly-under agents), while the process indicators were the operation indicators from the functional centers. In order to ascertain the performance indicators needed by the users, the methods such as unstructured interviews, structured interviews and records analysis were adopted by the project team. And also to determining the indicators, a systematic approach of KPI screening was used.

Television organizations were divided into two dimensions: the vertical dimension of organization-levels and the horizontal dimension of functions sectors, and the functionaries of each intersection were interviewed. Interviews focused on several aspects such as “problems concerned”, “measures taken”, “performance indicators”, “information needs”, “system requirements” and so on. In this way, if the organization has m departments and each department has n indicators, totally $3*m*n$ indicators could be obtained. According to the basic organization departments, the project team designed the targeted questionnaire in advance, gave systemic interview to the related staff, and then filled the designed electronic forms with the content obtained, so that the concerned collection of indicators could be captured.

By using this method, the project team collected hundreds of related indicators, which are used to evaluate the various departments of the organization, and then reduced these indicators to the five categories of factors: “finance”, “marketing and advertising”, “programs production and broadcasting”, “personnel” and “reflection from society”.

Although the idea of Balance Scorecard could be referred when inducing the performance indicators [8, 9], several KPIs which could reflect the particularity of the organization needed to be carefully chosen. For example, audience rating is a very important indicator when evaluating the quality of programs, but whether the programs are complied with the guidance of relevant state departments, as well as the comments from professionals in related fields are equally important. Again, the economic benefits generated by the program are very important, but CTV is not a profit-making organization, its advertising revenue from the centers could not be calculated using the general approach in enterprises. Hence some special problems were needed to be clarified, such as what factors should be included in the collection of KPIs, which are the higher-level indicators, which are lower-level indicators, what are the relationships between different indicators. Different users and departments had their own views and expressions. In order to verify a certain sector counterpart performance indicator, different opinions from several users were considered. A lot of research time was spent on understanding the significance of these indicators to business management, and the relationships between different indicators. Therefore, it was a iterative process that started from the collection of a large number of variables which were related to the few problems concerned by the top leaders. Then, the team went through the collation of all the variables and finally condensed them into a few models or KPIs.

For example, according to research and analysis, we found that most problems system users concerned were about the TV products (programs), such as the programmed production costs, the impact and the economic benefits generated. Therefore, the project team decided “the input and output of programs” as one of the

system’s KPIs. The calculation method of input-output of programs was: dividing input into financial investment and non-financial inputs, and dividing outputs into social benefits indicators and economic benefits indicators. This decomposition result of output indicators was shown in Fig. 1.

3.4 Data collection and collation

The CTV’s data comes from different data sources of many departments and sectors. In order to build the database of BPM system, to obtain integrated data with no conflicts, the process of analysis and collation of existing data was needed. The IT staff of project team had cleared the existing information systems and the data up first. After the investigation of the various business information systems, data were classified by different properties, such as organizations, related systems, the source of data and data formats, and also the various sources of data were cleared up too. After finishing lots of clearing-up-work, the project team has obtained all relevant data sources related to organizational performance, and data-tables were formed.

On this basis, IT personnel cleaned the existing data up in accordance with the significance and relevance of data. According to the organizational structure and program, data was divided into several dimensions: organizations, products, technology, broadcast, costs, advertising and so on, every dimension can be further decomposed to the final data items. For example, the organization dimension includes: the station, centers, departments, teams, individuals; the product dimension includes: channels, columns and program. Product data records content was associated with the production, including the time-length produced and/or the time-length bought of a program, and so on. Finally, several data properties such as the data-volume, the time dimension of data (such as per-month, per-quarter, and per-half-of-a-year), data storage (digital and electronic) and the formats. Through the above steps, the key performance indicators and the data sources of operational models were determined.

In the BPM system design, another important task is to design the model used in the data analysis. We can know from the definition of BPM that because of different enterprises had different requirements to performance analysis, it’s difficult to design a universal analysis model. Usually the models are developed according to the user’s industry characteristics, organizational structure and the user needs. So in the process of system research, the project team first studied the management problems and performance indicators users concerned, and then referred to the chosen KPIs and the data collection to consider how to design and implement the specific analysis models.

BPM analytical capacities bring the following benefits to performance management: 1) it can reveal the phenomenon which can’t be expressed by static indicators. For example, trend-forecast-model could analyze the development trend of performance indicators in a certain

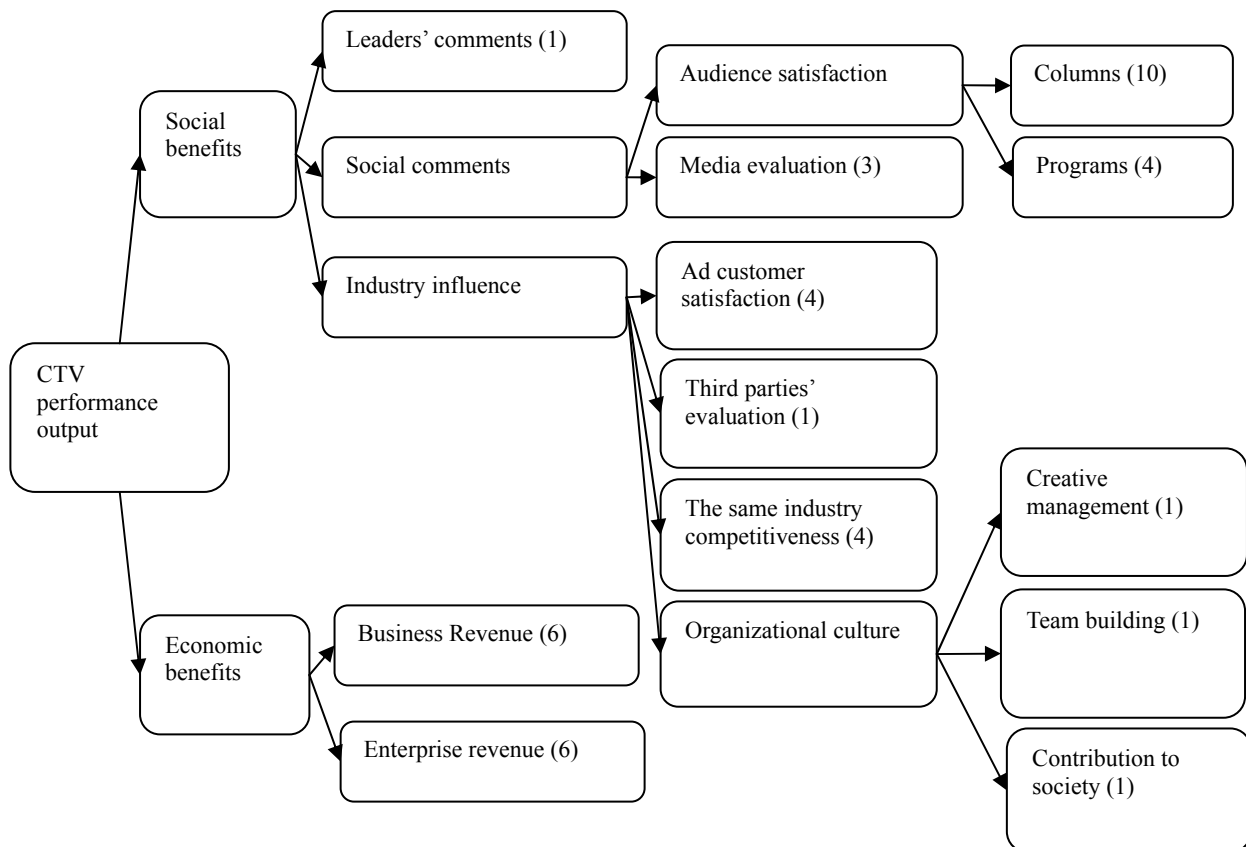


Fig.1 KPI development loop in BPM system development

period and forecast the possible problems. 2) Through structural analysis, comparative analysis can describe performance or issues which can't be expressed by a single indicator. For example, by comparing different performance indicators from several departments, the gap between them could be shown directly. 3) By using impact-analysis-model, the pivotal indicators that determined the KPI could be found out, also the major impacting factors in logical could be detected.

Take the mentioned business sector input and output indicators for example, the project team designed a series of operations model according to the corresponding data variables, which allows users to analyze the performance indicators directly, as well as the relationship between the indicators. The names and functions of the six models designed in the system were shown in Tab. 2.

3.5 User interface and digital dashboard design

There are two major considerations about the system interface design: 1) Use a simple intuitive way to provide user information. Generally users can obtain performance information through digital dashboard. When needing to do further downward mining or get detail information, user can open the corresponding data sheets in another window. 2) Try best to provide an integrated reporting environment by using uniform templates. After chosen a set of data users have the ability to customize their digital dashboard according to their own requirements. Such data demonstration approach enables users to present data from digital panel

to panel. User doesn't have to make separate information query by using multiple screens or multiple interfaces. The BPM project team completed the ultimate requirements analysis report basing on the research processes described above. Then the team built an analysis data collection for the system by analyzing the operational data. These results were delivered to the IT staff of the development team to develop databases, library of models and the digital instrument panel. Such a data demonstration approach enables users to present data from digital panel to panel. User doesn't have to make separate information query by using multiple screens or multiple interfaces.

4 The developing methodology of BPM systems

4.1 Systems development process

Based on the case study, we can identify the process of BPM system development and generalize the developing methodology of BPM systems. Though developing BPM systems in some places is similar to building ERP systems, they have different foundations [10]. First, the development of ERP based on the processes embed in software packages, by setting the inside parameters to customize a special enterprise system. BPM system development is mainly based on summarized enterprise data set. The system developer must understand the management goal for all hierarchical

Tab.2 The models of performance analysis

Name	Instructions	Examples
Structure analysis	Using pie-charts to express the revenue or cost structures of the various departments or business units (centers, channels, columns), by comparing the structures to shown the whole status of the organization.	Understand the structure characteristics of various columns or channels' cost of human resource, facilities and business in the process of cost analysis.
Sort analysis model	Using histograms, the horizontal axis represents various business departments or analysis units and the longitudinal axis represents the specified KPI; the system could make the synthesis ranking automatically. Users could set or adjust the weights of the indicators according to their significance.	By ranking the audience rating of columns to find out good columns as well as bad ones; so that notice could be drawn and the opportunities of improved methods could be get.
Relation analysis model	Use curve fitting plans to help users understand the correlation between performance indicators.	For example, use figure to express the relation between the columns' cost and ad incomes.
Impact analysis model	Using histograms, the horizontal axis represents the impacting factor while the vertical axis represents the impacted factor. The impact of each input could be determined by statistics.	For example: the different impacts of channels, period of time and types of programs to the advertisement incomes could be found.
Trend analysis model	Using broken line plans, the horizontal axis represents time dimension, such as month, quarter or year, and the vertical axis represents the development trend. By using time series analysis technology, the development trends could be shown.	The audience rating's changing statues of a certain column in a certain year. The model could help leaders to handle the future changes in a certain extent.
Warning analysis model	Using broken line plans with warning identifiers. User can set up the corresponding warning standards on the vertical axis to monitor the situation of performance indicators. Determine whether some issue exists by watching the development of indicators over time.	Monitor the use of budget; also could send warning information to the sector with a bad performance.

managers, and to design the key performance indicators for different functions. Second, there are few general BPM development tools since BPM applications are usually for special functional usage (such as budgeting, strategic finance, etc.). Finally, BPM systems contain a set of relatively new applications and analytical procedures. Though there are some DSS developing methodology can support BPM systems design [11, 12, 13], but companies still lack a complete methodology to guide the holistic BPM development process.

Our research group analyzed the real cases gathered from BPM applications provided by Hyperion and Oracle. Based on these real cases, we identified the common five phases involved with BPM systems development:

(1) System Background Analysis: Find out the competing strategy of the organization and the purpose of building BPM. By analyze the business process of the company, the organization structure, and the management goals the company, to clarify the users, scale and the overall structure of the system.

(2) Performance indicator analysis: Make a thorough discussion with the stakeholders of the

performance problems. Understand the demand of the performance management from different levels in the company. Collect the key performance indicators (KPI) based on the perspectives of the users. Define the KPI system, and the measurement method.

(3) BPM System design: This phase includes three aspects. First clarify the relationship among different performance management factors, and the KPIs. Second, design a new performance management information system based on enterprise data. According to different organization levels, to design the data manipulation models, such as data dimensions, data cubes, and data marts. And also design the corresponding performance analysis model. Design the BPM representation formats according to the KPIs, to design the dashboard, figures, report tables and other components.

(4) BPM System Implementation: Build the data warehouse and data marts and complete data extract, transform, clean, load, programming and multiple dimension database development.

(5) Applying and improvement: After the BPM system was applied, return to previous steps, to improve the system design by hearing the feedback of users [14].

4.2 Performance indicator design

Because there are different performance supervision methods and measurement methods in different organizations, KPI design must be conducted according to actual management circumstances. In the design phase, we can combine the two methods named “top down” and “bottom up”. As shown in Fig. 2, first we start from the strategic goal, the organization structure and the business character in the company, determine a few strategic indicators. Second, break down the indicators. Along with the descending of the organization level, collect other performance indicators related to the organization goal more broadly. Based on the understanding toward the KPIs in different departments, discuss with the critical issues from the grass roots level of the organization. Through the discussion, determine the KPIs in all levels of the organization. This is an ongoing knowledge sharing loop from volatilization to centralization. Along with the ascending of the discussion level, the problem scope should be smaller and smaller, until the group gets the most important KPI.

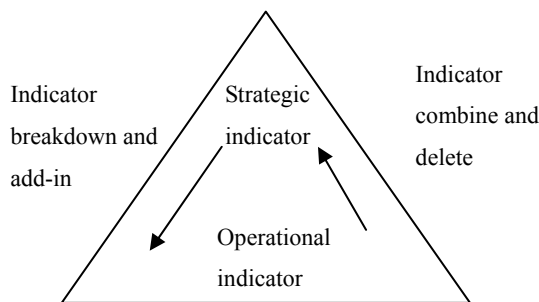


Fig.2 KPI development loop in BPM system development

In the KPI design phase, we should take the principles below.

(1)Conciseness: The level and the goal of the organization are the foundation of the definition and the measurement of KPI. Focusing on “the few indicators that the managers concerns most” is the key point of the KPI design. The indicators should be broken down according to the level of the organization. Although large numbers of indicators could give a whole view of the business performance, they would lead to a more complex system and the data analysis and operation could be too intricate. In a word, in the designing of KPI and related operational models, the indicators system should be concise and the number of the critical indicators in every aspect should be less than ten.

(2)Association: Virtually, the performance indicators are close related. We should pay more attention on the association of the indicators in the process of investigation. The investigators should have much knowledge about the business of the enterprise so that they could find out the corresponding relationship between the anfractuons businesses. The BPM system should avoid the isolation of the indicators by all means.

(3)Balance: Although the cost, the income and the

profit are important things that must be considered in the KPI design, some other indicators dose not related to finance cannot be ignored too. According to the concept of balanced score card, the project team should balance the financial indicators and non-financial indicators, inside indicators and outside indicators, indicators between different departments, the current profit and the future development. In addition, only depend on the indicators that computed by the system is not enough. It is necessary that the system provides the function for the users to make some changes according to actual circumstances, so that the users could make a balance between the objective data and the subjective decision.

In the BPM development process, the performance demand analysis phase is the most time consuming and fallibly phase. The developer must get into the company and work closely with the managers in every business department, in order to make sure that the KPI performance management method will be adopted properly. The discussion may refer to the performance indicators. Some general problems are shown in Tab. 4.

4.3 Team work in the development

BPM system development is a “social-technical” project involved with the knowledge form various domains. Some development methodology used in DSS and EIS could be referred in the BPM system project. The key successful issues are that the developer masters the concept of BPM and the IT staff and collaborate closely with the systems users. Fig. 3 shows the basic process of the system design. In the BPM system design, there are two different processes: the “BPM process” and the “IT process” [15]. The main mission of the “BPM process” is to define the performance indicators of the system [16], and to decide the mode in which the system provides support. The main mission of the “IT process” is to define the data set of the business system, to make the current information process mode and all the current information systems clear, and to consider how to develop the system by using some technology tools. Generally, there are two teams who are responsible for the two missions respectively. As a result, they should cooperate with and depend on each other to make the process smooth.

From the process we come to the conclusion that the communication and the cooperation between members in the team determine the speed and the quality of the project. The three conclusions from the case are:

(1) There should be definite division of work in the project team. People who are responsible for BPM should be up on the organization structure and the business process as soon as possible. Only have an enough understand could the team communicate with professionals in other related department (such as: financial department, marketing department) deeply. On the other hand, people who are responsible for IT should master the information systems condition in the organization quickly, and handle the BPM realization tools.

Tab.4 The general problems of the performance demand analysis phase

Type	Problem
Performance indicator	What should the indicator be named? How is it be used and measured in the organization now?
Purpose and meanings	Why should we measure this indicator? What about its importance? What about the purpose of the indicator? What behaviors dose the indicator support? Are there any other behaviors affect this indicator? Is it possible to lead to negative behaviors?
Measurement formula	How can we measure the performance? Can it be defined in mathematical version? What kind of data dose the measure need? (collect/ check period) Is the data accurate and timely enough?
Correlation variable	What variables have close relationship with this measure method? What measure methods have close relationship with this method?
Users (owners)	Who are responsible for the measure of the performance (name, function or outside organizations)? Who are responsible for the data collection, correction and analysis?
Function	How does the method owner use the data? What kind of analyze could the user do to the result of the performance? What kind of behavior could the user take?

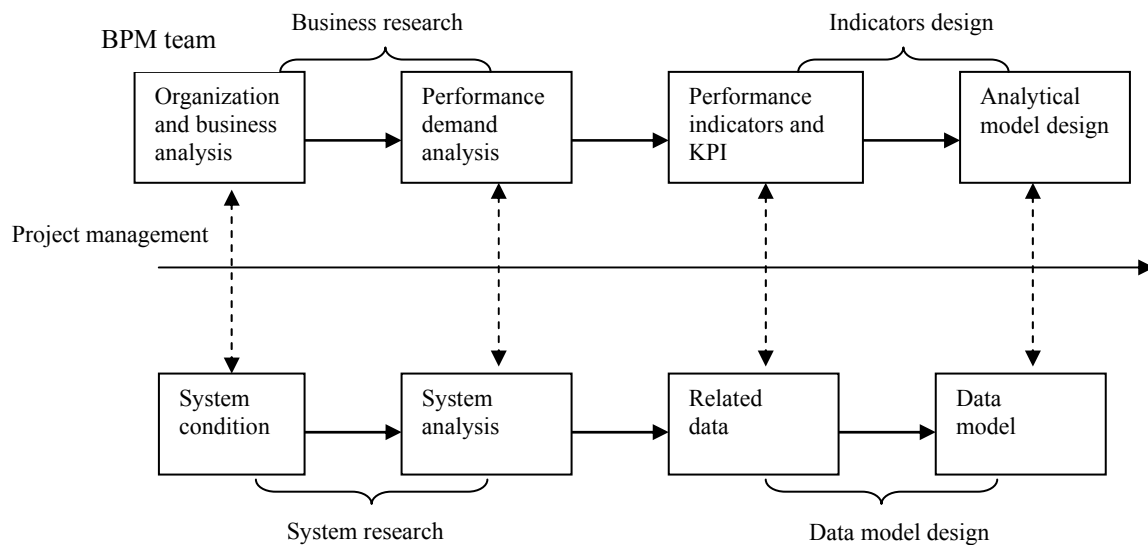


Fig.3 The basis process of BPM systems design

(2) On one hand, the two kinds of members should push their work. On the other hand, they should try to avoid the lack of communication. The project manager is expected to have strong leadership, and try to integrate the two works. Otherwise, members may pass the buck to each other.

(3) Virtually, BPM demanding analysis is a cycle contains three steps: “propose a hypothesis”, “feedback”, “modify the hypothesis”. According to the survey result, the project team proposes a hypothesis. Let the uses give the feedback meanwhile. It is advisable to get the feedback from the executive managers. Prototype

method and graphical interface method could be used in this phase. The team could show the work mode and capacity to the users through demonstration project.

The realization of BPM is different from the development of traditional ERP systems. Although there are some ready-made BPM development tools, there isn't any current BPM systems crust that could be used directly. Generally, the development tool is to be used in one aspect in BPM cycle from strategic level to operational level [17] (e.g. strategic planning, corporate budgeting, financial reporting). To realize a system satisfying customers' all demands in performance management need the integration of different tools [18]. Therefore, there should be a long-term plan for research on BPM systems realization beyond the performance measurement paradigm [19]. Under the guidance of the plan, start from specific goals, develop the performance indicator system and the management platform that suit to the customer gradually [20]. In this process, the corporation managers grind in with the IT professions according to the model above is absolutely necessary.

5 Conclusions

This paper discusses the concept and the methodology for developing BPM systems. Based on the real case study, the research analyzed the characteristics and critical issues of the design and the implementing of BPM systems. According to the case discussed in the paper, BPM systems design is the application process of the socio-technical system development methodology. It is necessary that the various professions cooperate closely with each other. A systematic concept framework and methodology of BPM development will guide the team when conducting the project. It will avoid pitfalls in the collaboration processes and improve the success rate of the systems. The methods and frameworks described in the paper depend mainly on the researchers' BPM development experience and understanding. The methodology could be extended in the future research on BPM systems development. As there will be more cases studied in the future work, the insights and experiences accumulated from the practices and research works will help develop an integrated BPM systems design methodology.

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